

Synthesis of New Chiral *cis*-3-Hydroxyazetidines and Their Application in Diethylzinc Addition to Aldehydes

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ABSTRACT A new series of chiral *cis*-3-hydroxyazetidines have been prepared from (*R*)-1-phenylethylamine. They have excellent catalytic activities and enantiomeric selectivities in asymmetric addition of diethylzinc to aromatic aldehydes. *Chirality* 19:802–808, 2007. © 2007 Wiley-Liss, Inc.

KEY WORDS: *cis*-3-hydroxyazetidines; chiral auxiliary; synthesis; application; asymmetric diethylzinc addition reaction

INTRODUCTION

The stereoselective addition of organometallics to one of the two heterotopic faces of a carbonyl group has been extensively studied.^{1–4} Among them, the enantioselective addition of organozinc to carbonyl compounds is of particular importance in organic synthesis, because the products, chiral alcohols, represent valuable structural units found in many natural products and they are readily functionalized.⁵ Since the pioneering work of Oguni in 1984,⁶ there has been extensive research into the diethylzinc addition to benzaldehyde, and it is now regarded as one of the benchmark reactions for understanding the catalytic potential of new ligands.

Although a variety of ligands, such as chiral amino alcohols, amino thiols, amino disulfides, amino diselenides, diamines, or diols have been studied,^{7–11} the development of new ligands for asymmetric addition of diethylzinc to aldehydes is a desirable goal. Chiral azetidines have been extensively used as a chiral auxiliary in recent years,¹² but only few studies have focused on the applications of its derivatives as chiral catalysts.^{13–24} Chiral azetidines that can effectively catalyze the diethylzinc addition with high levels of enantioselectivity are scarcely reported,^{20–24} and to our knowledge, no example of chiral hydroxyazetidine (hydroxyl group seated on the azetidine ring) for enantioselective catalysis has been described. We report herein the synthesis of a new series of chiral *cis*-3-hydroxyazetidines and their application in asymmetric addition of diethylzinc to aldehydes.

EXPERIMENTAL SECTION

All chemicals were purchased from Aldrich Chemical Co. and Beijing Chemical Co. used as received unless otherwise noted. Infrared spectra were obtained on an Avatar 360 Fourier transform spectrometer. Single-crystal X-ray diffraction measurements were carried out on a Bruker CCD diffractometer. ¹H and ¹³C NMR spectra were recorded on Bruker AV-300, AVQ-400, and AV-500 spectrometers. All chemical shifts were reported in δ units with reference to the residual protons of the deuterated

solvents for proton and carbon chemical shifts. Melting points were measured on an X-6 melting point apparatus and were uncorrected. Optical rotations were measured on a PerkinElmer 343 polarimeter. Elemental analyses were performed on a Vario EL elemental analyzer. Gas chromatography was conducted on Varian CP-3800 series with FID detector using chiral column SGE β -CYCLODEX, 25 m $\times$ 0.25 mm I.D. Retention time was given in minutes.

General Procedure for the Preparation of *cis*-3-Hydroxyazetidines from (*R*)-1-Phenylethylamine

The synthesis of 1-(1'-phenylethyl)-2-phenyl-3-hydroxyazetidine (**L1**) is representative. Anhydrous MgSO₄ (4.8 g, 40 mmol) was added to a CH₂Cl₂ (60 ml) solution of benzaldehyde (2.14 g, 20 mmol) and (*R*)-1-phenylethylamine (2.42 g, 20 mmol) at room temperature. After this solution was stirred at room temperature for 12 h, MgSO₄ was filtered off. To the resulting solution, triethylamine (4.04 g, 40 mmol) and acetoxyacetyl chloride (5.46 g, 40 mmol) were slowly added with stirring at -10°C . After the solution was stirred for 24 h at room temperature, triethylamine hydrochloride was filtered off, and the filtrate was washed with saturated NaHCO₃ (25 ml $\times$ 2). Drying over anhydrous MgSO₄, solvent was removed to give a crude mixture (**2a** and **2b**). The mixture was dissolved in 40 ml of dry THF, then the solution was added to a suspension of aluminum chloride (10.68 g, 80 mmol) and lithium aluminum hydride (3.04 g, 80 mmol) in 150 ml of dry THF with stirring at reflux. After this mixture was refluxed for 3 h, then cooled to 0°C , and 40 ml of water was carefully added. The water phase was extracted with dichloromethane (50 ml $\times$ 3). The organic layer was dried over anhydrous

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MgSO₄, then solvent was removed to give a crude mixture, 1-(1'-phenylethyl)-2-phenyl-3-hydroxyazetidine (**L1a** and **L1b**), which was further purified by chromatography (hexane/ethyl acetate = 5/1) to afford pure compounds (1'*R*,2*R*,3*R*)-1-(1'-phenylethyl)-2-phenyl-3-hydroxyazetidine (**L1a**; 2.18 g, 43%) and (1'*R*,2*S*,3*S*)-1-(1'-phenylethyl)-2-phenyl-3-hydroxyazetidine (**L1b**; 1.06 g, 21%), total yield: 3.24 g, 64%.

(1'*R*,2*R*,3*R*)-1-(1'-phenylethyl)-2-phenyl-3-hydroxyazetidine (L1a)

Colorless crystals, mp 136–138°C; $[\alpha]_D^{20} = -188.0$ (*c* 0.67, CH₂Cl₂); ¹H NMR (300 MHz, CDCl₃): δ 1.30 (d, *J* = 6.6 Hz, 3H), 1.71 (d, *J* = 5.7 Hz, 1H), 3.29 (m, 1H), 3.47 (m, 2H), 4.23 (d, *J* = 6.0 Hz, 1H), 4.38 (q, *J* = 5.7 Hz, 1H), 6.99–7.07 (m, 3H), 7.09–7.26 (m, 7H) ppm; ¹³C NMR (75 MHz, CDCl₃): δ 19.5, 58.6, 66.0, 66.7, 72.0, 127.0, 127.1, 127.6, 127.7, 128.0, 136.9, 142.1 ppm; IR (KBr): 3464, 2963, 1602 cm⁻¹; MS (EI): *m/z* 253 (M⁺); Anal. Calcd for C₁₇H₁₉NO: C, 80.6; H, 7.56; N, 5.53. Found: C, 80.3; H, 7.50; N, 5.43.

(1'*R*,2*S*,3*S*)-1-(1'-phenylethyl)-2-phenyl-3-hydroxyazetidine (L1b)

Colorless crystals, mp 108–110°C; $[\alpha]_D^{20} = +116.3$ (*c* 0.32, CH₂Cl₂); ¹H NMR (400 MHz, CDCl₃): δ 1.06 (d, *J* = 6.4 Hz, 3H), 1.68 (s, 1H), 3.04 (m, 1H), 3.16 (d, *J* = 8.8 Hz, 1H), 3.53 (q, *J* = 6.4 Hz, 1H), 4.42 (s, 2H), 7.27–7.47 (m, 8H), 7.63 (d, *J* = 7.6 Hz, 2H) ppm; ¹³C NMR (100 MHz, CDCl₃): δ 22.9, 59.0, 65.8, 67.9, 72.2, 127.0, 127.2, 127.6, 127.8, 128.3, 128.4, 138.4, 143.7 ppm; IR (KBr): 3412, 2973, 1605 cm⁻¹; MS (EI): *m/z* 253 (M⁺); Anal. Calcd for C₁₇H₁₉NO: C, 80.6; H, 7.56; N, 5.53. Found: C, 80.5; H, 7.53; N, 5.53.

(1'*R*,2*R*,3*R*)-1-(1'-phenylethyl)-2-(2-chlorophenyl)-3-hydroxyazetidine (L2a)

Colorless crystals, mp 50–52°C; $[\alpha]_D^{20} = -112.8$ (*c* 0.19, CH₂Cl₂); ¹H NMR (500 MHz, CDCl₃): δ 1.40 (d, *J* = 5.5 Hz, 3H), 1.52 (d, *J* = 5.0 Hz, 1H), 3.41 (m, 1H), 3.51 (m, 2H), 4.63 (q, *J* = 5.5 Hz, 1H), 4.69 (br, s, 1H), 7.05–7.16 (m, 5H), 7.28 (m, 3H), 7.49 (br, s, 1H) ppm; ¹³C NMR (125 MHz, CDCl₃): δ 18.6, 58.6, 65.3, 66.4, 69.2, 125.9, 127.2, 127.9, 128.0, 128.4, 130.7, 131.7, 134.6, 142.0 ppm; IR (KBr): 3438, 2970, 1630 cm⁻¹; MS (EI): *m/z* 286 (M⁺ – 1); Anal. Calcd for C₁₇H₁₈ClNO: C, 71.0; H, 6.30; N, 4.87. Found: C, 70.9; H, 6.60; N, 4.79.

(1'*R*,2*S*,3*S*)-1-(1'-phenylethyl)-2-(2-chlorophenyl)-3-hydroxyazetidine (L2b)

Colorless crystals, mp 92–94°C; $[\alpha]_D^{20} = +140.4$ (*c* 0.29, CH₂Cl₂); ¹H NMR (500 MHz, CDCl₃): δ 1.10 (d, *J* = 6.5 Hz, 3H), 1.54 (d, *J* = 6.5 Hz, 1H), 3.12 (dd, *J* = 9.5, 6.5 Hz, 1H), 3.19 (d, *J* = 9.5 Hz, 1H), 3.58 (q, *J* = 6.5 Hz, 1H), 4.65 (q, *J* = 5.5 Hz, 1H), 4.73 (d, *J* = 6.0 Hz, 1H), 7.27–7.46 (m, 8H), 8.13 (d, *J* = 7.0 Hz, 1H) ppm; ¹³C NMR (125 MHz, CDCl₃): δ 22.9, 59.3, 65.0, 68.1, 69.7, 126.6, 127.2, 127.3, 128.4, 128.6, 129.0, 130.8, 132.0, 136.3, 143.7 ppm; IR (KBr): 3442, 2965 cm⁻¹; MS (EI): *m/z* 286 (M⁺ – 1); Anal.

Calcd for C₁₇H₁₈ClNO: C, 71.0; H, 6.30; N, 4.87. Found: C, 70.9; H, 6.39; N, 4.64.

(1'*R*,2*R*,3*R*)-1-(1'-phenylethyl)-2-(3-chlorophenyl)-3-hydroxyazetidine (L3a)

Colorless crystals, mp 89–91°C; $[\alpha]_D^{20} = -200.2$ (*c* 0.44, CH₂Cl₂); ¹H NMR (500 MHz, CDCl₃): δ 1.35 (d, *J* = 6.5 Hz, 3H), 1.65 (br, s, 1H), 3.32 (m, 1H), 3.46 (q, *J* = 6.2 Hz, 1H), 3.50 (d, *J* = 8.2 Hz, 1H), 4.20 (d, *J* = 5.2 Hz, 1H), 4.43 (q, *J* = 5.6 Hz, 1H), 7.03–7.10 (m, 6H), 7.18–7.21 (m, 3H) ppm; ¹³C NMR (125 MHz, CDCl₃): δ 19.5, 58.8, 66.1, 67.1, 71.7, 126.0, 127.1, 127.3, 127.8, 128.1, 128.3, 128.9, 133.7, 139.3, 141.9 ppm; IR (KBr): 3437, 2975 1601 cm⁻¹; MS (EI): *m/z* 287 (M⁺); Anal. Calcd for C₁₇H₁₈ClNO: C, 71.0; H, 6.30; N, 4.87. Found: C, 70.9; H, 6.45; N, 4.83.

(1'*R*,2*S*,3*S*)-1-(1'-phenylethyl)-2-(3-chlorophenyl)-3-hydroxyazetidine (L3b)

Colorless crystals, mp 64–66°C; $[\alpha]_D^{20} = +93.2$ (*c* 0.31, CH₂Cl₂); ¹H NMR (500 MHz, CDCl₃): δ 1.05 (d, *J* = 6.5 Hz, 3H), 1.61 (d, *J* = 6.5 Hz, 1H), 3.05 (m, 1H), 3.15 (d, *J* = 9.0 Hz, 1H), 3.52 (q, *J* = 6.5 Hz, 1H), 4.36 (d, *J* = 6.0 Hz, 1H), 4.43 (q, *J* = 6.0 Hz, 1H), 7.28–7.50 (m, 8H), 7.64 (s, 1H) ppm; ¹³C NMR (125 MHz, CDCl₃): δ 22.7, 59.0, 65.9, 67.8, 71.8, 126.0, 127.1, 127.2, 127.8, 128.1, 128.4, 129.6, 134.5, 140.9, 143.4 ppm; IR (KBr): 3423, 2966 1600 cm⁻¹; MS (EI): *m/z* 286 (M⁺ – 1); Anal. Calcd for C₁₇H₁₈ClNO: C, 71.0; H, 6.30; N, 4.87. Found: C, 70.9; H, 6.64; N, 4.81.

(1'*R*,2*R*,3*R*)-1-(1'-phenylethyl)-2-(4-chlorophenyl)-3-hydroxyazetidine (L4a)

Colorless crystals, mp 105–107°C; $[\alpha]_D^{20} = -204.2$ (*c* 0.43, CH₂Cl₂); ¹H NMR (500 MHz, CDCl₃): δ 1.34 (d, *J* = 6.5 Hz, 3H), 1.63 (br, s, 1H), 3.33 (m, 1H), 3.46 (d, *J* = 6.0 Hz, 1H), 3.51 (d, *J* = 8.5 Hz, 1H), 4.20 (d, *J* = 5.5 Hz, 1H), 4.42 (d, *J* = 5.5 Hz, 1H), 7.04–7.08 (m, 3H), 7.12–7.17 (m, 6H) ppm; ¹³C NMR (125 MHz, CDCl₃): δ 19.7, 58.9, 66.0, 67.1, 71.6, 127.2, 127.8, 127.9, 128.0, 129.5, 132.7, 135.7, 142.0. IR (KBr): 3461, 2943, 1598 cm⁻¹; MS (EI): *m/z* 286 (M⁺ – 1); Anal. Calcd for C₁₇H₁₈ClNO: C, 71.0; H, 6.30; N, 4.87; Found: C, 70.9; H, 6.66; N, 4.79.

(1'*R*,2*S*,3*S*)-1-(1'-phenylethyl)-2-(4-chlorophenyl)-3-hydroxyazetidine (L4b)

Colorless crystals, mp 110–112°C; $[\alpha]_D^{20} = +99.7$ (*c* 0.30, CH₂Cl₂); ¹H NMR (500 MHz, CDCl₃): δ 1.03 (d, *J* = 6.5 Hz, 3H), 1.56 (d, *J* = 6.5 Hz, 1H), 3.04 (m, 1H), 3.14 (d, *J* = 9.0 Hz, 1H), 3.51 (q, *J* = 6.5 Hz, 1H), 4.36 (d, *J* = 6.0 Hz, 1H), 4.42 (q, *J* = 6.0 Hz, 1H), 7.27–7.42 (m, 7H), 7.57 (d, *J* = 8.0 Hz, 2H) ppm; ¹³C NMR (125 MHz, CDCl₃): δ 22.9, 59.0, 65.9, 67.9, 71.7, 127.1, 127.2, 128.4, 128.6, 129.3, 133.4, 137.1, 143.5 ppm; IR (KBr): 3432, 2966 cm⁻¹; MS (EI): *m/z* 287 (M⁺); Anal. Calcd for C₁₇H₁₈ClNO: C, 71.0; H, 6.30; N, 4.87. Found: C, 70.9; H, 6.21; N, 4.75.

(1'R,2R,3R)-1-(1'-phenylethyl)-2-(2-methoxyphenyl)-3-hydroxyazetidine (L5a)

Colorless oil, $[\alpha]_{\text{D}}^{20} = -100.0$ (c 0.17, CH_2Cl_2); ^1H NMR (500 MHz, CDCl_3): δ 1.43 (d, $J = 5.4$ Hz, 3H), 1.68 (br, s, 1H), 3.42 (br, s, 1H), 3.51 (br, s, 1H), 3.57 (br, s, 1H), 3.79 (s, 3H), 4.55 (q, $J = 5.9$ Hz, 1H), 4.73 (br, s, 1H), 6.73 (d, $J = 8.1$ Hz, 1H), 6.82 (t, $J = 7.4$ Hz, 1H), 7.10–7.19 (m, 4H), 7.32 (m, 3H); ^{13}C NMR (125 MHz, CDCl_3): δ 18.7, 55.4, 58.8, 65.7, 66.2, 66.9, 109.6, 120.0, 125.2, 127.1, 127.8, 127.9, 128.0, 129.8, 142.6, 156.1 ppm; IR (neat): 3415, 2965, 1601 cm^{-1} ; MS (EI): m/z 283 (M^+); Anal. Calcd for $\text{C}_{18}\text{H}_{21}\text{NO}_2$: C, 76.3; H, 7.47; N, 4.94. Found: C, 76.3; H, 7.47; N, 4.78.

(1'R,2S,3S)-1-(1'-phenylethyl)-2-(2-methoxyphenyl)-3-hydroxyazetidine (L5b)

Colorless oil, $[\alpha]_{\text{D}}^{20} = +85.5$ (c 0.11, CH_2Cl_2); ^1H NMR (500 MHz, CDCl_3): δ 1.14 (d, $J = 6.4$ Hz, 3H), 1.65 (d, $J = 6.3$ Hz, 1H), 3.10 (m, 1H), 3.21 (d, $J = 9.2$ Hz, 1H), 3.57 (q, $J = 6.4$ Hz, 1H), 3.92 (s, 3H), 4.55 (q, $J = 6.2$ Hz, 1H), 4.77 (d, $J = 5.8$ Hz, 1H), 6.95 (d, $J = 8.1$ Hz, 1H), 7.14 (t, $J = 7.2$ Hz, 1H), 7.03–7.50 (m, 6H), 8.04 (d, $J = 7.2$ Hz, 1H) ppm; ^{13}C NMR (125 MHz, CDCl_3): δ 23.1, 55.4, 59.8, 65.3, 67.5, 68.4, 109.8, 120.5, 126.7, 127.0, 127.2, 127.3, 128.4, 129.8, 144.1, 156.3 ppm; IR (neat): 3421, 2965, 1600 cm^{-1} ; MS (EI): m/z 283 (M^+); Anal. Calcd for $\text{C}_{18}\text{H}_{21}\text{NO}_2$: C, 76.3; H, 7.47; N, 4.94. Found: C, 76.4; H, 7.81; N, 4.67.

(1'R,2R,3R)-1-(1'-phenylethyl)-2-(3-methoxyphenyl)-3-hydroxyazetidine (L6a)

Colorless crystals, mp 75–77°C; $[\alpha]_{\text{D}}^{20} = -199.3$ (c 0.43, CH_2Cl_2); ^1H NMR (500 MHz, CDCl_3): δ 1.35 (d, $J = 6.4$ Hz, 3H), 1.71 (br, s, 1H), 3.31 (m, 1H), 3.49 (m, 2H), 3.77 (s, 3H), 4.28 (d, $J = 5.2$ Hz, 1H), 4.40 (d, $J = 4.9$ Hz, 1H), 6.67 (m, 1H), 6.77 (m, 2H), 7.05–7.11 (m, 4H), 7.22 (d, $J = 7.0$ Hz, 2H) ppm; ^{13}C NMR (125 MHz, CDCl_3): δ 19.4, 55.2, 58.6, 66.0, 66.4, 72.0, 113.1, 113.2, 120.1, 127.1, 127.7, 128.1, 128.8, 138.7, 142.2, 159.3 ppm; IR (KBr): 3442, 2996, 1583 cm^{-1} ; MS (EI): m/z 283 (M^+); Anal. Calcd for $\text{C}_{18}\text{H}_{21}\text{NO}_2$: C, 76.3; H, 7.47; N, 4.94. Found: C, 76.3; H, 7.77; N, 4.90.

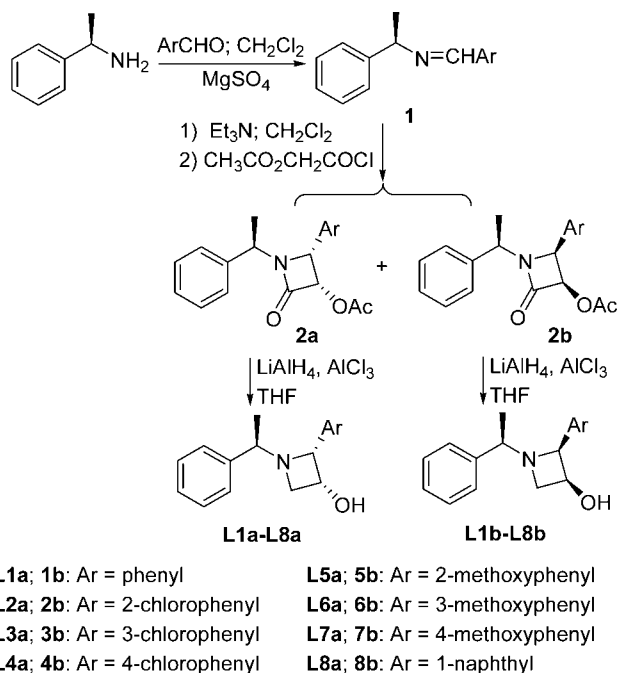
(1'R,2S,3S)-1-(1'-phenylethyl)-2-(3-methoxyphenyl)-3-hydroxyazetidine (L6b)

Colorless oil, $[\alpha]_{\text{D}}^{20} = +91.7$ (c 0.22, CH_2Cl_2); ^1H NMR (500 MHz, CDCl_3): δ 1.08 (d, $J = 6.2$ Hz, 3H), 1.70 (br, s, 1H), 3.02 (m, 1H), 3.15 (d, $J = 8.9$ Hz, 1H), 3.52 (d, $J = 6.2$ Hz, 1H), 3.90 (s, 3H), 4.40 (s, 2H), 6.98 (dd, $J = 8.0$, 1.6 Hz, 1H), 7.17–7.44 (m, 8H) ppm; ^{13}C NMR (125 MHz, CDCl_3): δ 22.9, 55.3, 59.1, 65.9, 67.9, 72.2, 113.0, 113.6, 120.0, 127.1, 127.3, 128.4, 129.5, 140.2, 143.7, 159.9 ppm; IR (neat): 3426, 2966, 1600 cm^{-1} ; MS (EI): m/z 283 (M^+); Anal. Calcd for $\text{C}_{18}\text{H}_{21}\text{NO}_2$: C, 76.3; H, 7.47; N, 4.94. Found: C, 76.4; H, 7.78; N, 4.90.

(1'R,2R,3R)-1-(1'-phenylethyl)-2-(4-methoxyphenyl)-3-hydroxyazetidine (L7a)

Colorless crystals, mp 98–100°C; $[\alpha]_{\text{D}}^{20} = -206.3$ (c 0.35, CH_2Cl_2); ^1H NMR (400 MHz, CDCl_3): δ 1.32 (d, $J = 6.8$ Hz, 3H), 1.72 (br, s, 1H), 3.28 (m, 1H), 3.49 (m, 2H),

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Scheme 1. Synthesis of chiral *cis*-3-hydroxyazetidines.

3.76 (s, 3H), 4.21 (d, $J = 6.0$ Hz, 1H), 4.37 (br, s, 1H), 6.72 (d, $J = 8.8$ Hz, 2H), 7.01–7.20 (m, 7H) ppm; ^{13}C NMR (100 MHz, CDCl_3): δ 19.6, 55.2, 58.5, 66.0, 66.6, 71.4, 113.2, 127.0, 127.6, 128.0, 129.0, 129.2, 142.2, 158.6 ppm; IR (KBr): 3473, 2938, 1586 cm^{-1} ; MS (EI): m/z 283 (M^+); Anal. Calcd for $\text{C}_{18}\text{H}_{21}\text{NO}_2$: C, 76.3; H, 7.47; N, 4.94. Found: C, 76.1; H, 7.37; N, 4.74.

(1'R,2S,3S)-1-(1'-phenylethyl)-2-(4-methoxyphenyl)-3-hydroxyazetidine (L7b)

Colorless crystals, mp 81–83°C; $[\alpha]_{\text{D}}^{20} = +97.1$ (c 0.31, CH_2Cl_2); ^1H NMR (500 MHz, CDCl_3): δ 1.04 (d, $J = 5.9$ Hz, 3H), 1.63 (br, s, 1H), 3.00 (br, s, 1H), 3.13 (d, $J = 8.7$ Hz, 1H), 3.51 (d, $J = 5.8$ Hz, 1H), 3.86 (s, 3H), 4.37 (s, 2H), 6.99 (d, $J = 8.4$ Hz, 2H), 7.35–7.43 (m, 5H), 7.56 (d, $J = 7.8$ Hz, 2H) ppm; ^{13}C NMR (125 MHz, CDCl_3): δ 22.9, 55.3, 58.9, 65.9, 68.0, 71.9, 113.9, 127.1, 127.3, 128.4, 129.1, 130.3, 143.8, 159.2 ppm; IR (KBr): 3511, 3020, 1608 cm^{-1} ; MS (EI): m/z 283 (M^+); Anal. Calcd for $\text{C}_{18}\text{H}_{21}\text{NO}_2$: C, 76.3; H, 7.47; N, 4.94. Found: C, 76.3; H, 7.44; N, 4.91.

(1'R,2R,3R)-1-(1'-phenylethyl)-2-(1'-naphthyl)-3-hydroxyazetidine (L8a)

Colorless crystals, mp 107–109°C; $[\alpha]_{\text{D}}^{20} = -100.6$ (c 0.53, CH_2Cl_2); ^1H NMR (500 MHz, CDCl_3): δ 1.46 (d, $J = 6.4$ Hz, 3H), 1.54 (d, $J = 6.6$ Hz, 1H), 3.53 (m, 2H), 3.65 (d, $J = 6.4$ Hz, 1H), 4.72 (d, $J = 6.0$ Hz, 1H), 5.12 (d, $J = 5.4$ Hz, 1H), 7.04 (d, $J = 7.2$ Hz, 1H), 7.10 (m, 2H), 7.30 (m, 3H), 7.47 (m, 2H), 7.64 (d, $J = 8.2$ Hz, 1H), 7.69 (d, $J = 6.8$ Hz, 1H), 7.82–7.87 (m, 2H) ppm; ^{13}C NMR (100 MHz, CDCl_3): δ 18.4, 58.7, 65.9, 66.1, 69.1, 122.2, 125.1, 125.4, 126.0, 126.7, 127.2, 127.4, 127.9, 128.0, 128.8, 130.4, 131.9, 133.3, 142.3 ppm; IR (KBr): 3565, 2935, 1595 cm^{-1} ;

TABLE 1. Synthesis and yields of (1'*R*,2*R*,3*R*)-2-Aryl-3-hydroxyazetidines (**La**) and (1'*R*,2*S*,3*S*)-2-Aryl-3-hydroxyazetidines (**Lb**) from (*R*)-1-phenylethylamine

Compound	Ar	Yield of La (%)	Yield of Lb (%)	Total yield (%)
L1a; 1b	Ph	43	21	64
L2a; 2b	2-ClC ₆ H ₄	48	22	70
L3a; 3b	3-ClC ₆ H ₄	39	22	61
L4a; 4b	4-ClC ₆ H ₄	38	24	62
L5a; 5b	2-MeOC ₆ H ₄	45	28	73
L6a; 6b	3-MeOC ₆ H ₄	36	18	54
L7a; 7b	4-MeOC ₆ H ₄	40	19	59
L8a; 8b	1-naphthyl	45	33	78

MS (EI): *m/z* 303 (*M*⁺); Anal. Calcd for C₂₁H₂₁NO: C, 83.1; H, 6.98; N, 4.62. Found: C, 82.9; H, 7.33; N, 4.64.

(1'*R*,2*S*,3*S*)-1-(1'-phenylethyl)-2-(1'-naphthyl)-3-hydroxyazetidine (L8b**)**

Colorless oil, [α]_D²⁰ = +116.0 (*c* 0.28, CH₂Cl₂); ¹H NMR (500 MHz, CDCl₃): δ 1.15 (d, *J* = 6.2 Hz, 3H), 1.57 (br, s, 1H), 3.21 (m, 1H), 3.27 (d, *J* = 9.0 Hz, 1H), 3.65 (d, *J* = 6.3 Hz, 1H), 4.74 (d, *J* = 5.8 Hz, 1H), 5.13 (d, *J* = 5.4 Hz, 1H), 7.31 (m, 1H), 7.40 (t, *J* = 7.5 Hz, 2H), 7.50–7.65 (m, 5H), 7.86 (d, *J* = 8.1 Hz, 1H), 7.97 (m, 2H), 8.31 (d, *J* = 6.5 Hz, 1H) ppm; ¹³C NMR (125 MHz, CDCl₃): δ 23.0, 59.8, 65.7, 68.4, 69.6, 122.3, 125.6, 125.8, 126.3, 126.9, 127.2, 127.4, 127.9, 128.5, 129.0, 130.6, 133.6, 143.7 ppm; IR (neat): 3409, 3059, 1596 cm⁻¹; MS (EI): *m/z* 303 (*M*⁺); Anal. Calcd for C₂₁H₂₁NO: C, 83.1; H, 6.98; N, 4.62. Found: C, 82.8; H, 7.23; N, 4.54.

X-ray Crystal Structure Data for **L6a and **L7b****

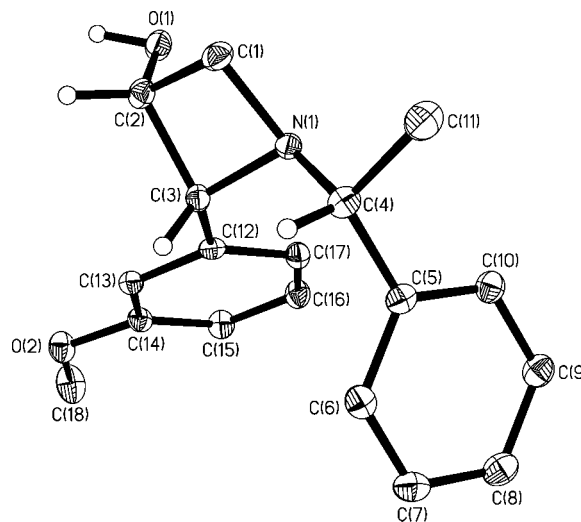
Compound **L6a**: C₁₈H₂₁NO₂, *M_r* = 283.36, monoclinic, *P*2₁, *a* = 9.1306(12) Å, *b* = 7.0797(9) Å, *c* = 12.5002(14) Å; β = 95.645(8)°; *V* = 804.12(17) Å³; 3.28° < 2 θ < 55.68°; ρ_{calc} = 1.170 g cm⁻³, *Z* = 2, *F*(000) = 304; Final *R*₁ = 0.0499, *wR*₂ = 0.1144, all data, *R*₁ = 0.0625, *wR*₂ = 0.1217.

Compound **L7b**: C₁₈H₂₁NO₂, *M_r* = 283.36, monoclinic, *P*2₁, *a* = 12.226(3) Å, *b* = 5.8444(16) Å, *c* = 12.296(3) Å; β = 114.222(4)°; *V* = 801.2(4) Å³; 3.64° < 2 θ < 52.86°; ρ_{calc} = 1.175 g cm⁻³, *Z* = 2, *F*(000) = 304; Final *R*₁ = 0.0396, *wR*₂ = 0.0827, all data, *R*₁ = 0.0737, *wR*₂ = 0.0973.*

General Procedure for Diethylzinc Addition to Aldehydes

The chiral ligand (0.1 mmol) was dissolved in toluene (3 ml) at room temperature, and diethylzinc (2.0 ml of 1.5 M of toluene solution, 3.0 mmol) was injected. After the mixture was stirred for 20 min, benzaldehyde (0.1 g, 1 mmol) was added dropwise via a syringe, and the mixture stirred for the corresponding reaction time under N₂. The reaction was quenched by the addition of a saturated aqueous solution of NH₄Cl (10 ml). The mixture was then extracted with Et₂O (15 ml $\times$ 3). The combined organic phases

were dried over anhydrous Na₂SO₄ and concentrated in vacuo. The crude products were purified by flash column chromatography (hexane/EtOAc = 4/1). The ee values of the alcohol products were determined by GC analysis on a chiral cyclodextrin capillary column. Conditions of GC analyses: N₂ as carrier gas (1.5 ml/min). For 1-phenylpropanol: DET = 280°C, INJ = 240°C, OVEN = 120°C, *t_R* = 20.0 min for (*R*) and *t_R* = 20.3 min for (*S*). For 1-(2-chlorophenyl)propanol: DET = 300°C, INJ = 300°C, OVEN = 150°C, *t_R* = 16.0 min for (*R*) and *t_R* = 16.2 min for (*S*). For 1-(3-chlorophenyl)propanol: DET = 280°C, INJ = 280°C, OVEN = 130°C, *t_R* = 48.0 min for (*R*) and *t_R* = 48.6 min for (*S*). For 1-(4-chlorophenyl)propanol: DET = 300°C, INJ = 300°C, OVEN = 150°C, *t_R* = 19.8 min for (*R*) and *t_R* = 20.2 min for (*S*). For 1-(2-methoxyphenyl)propanol: DET = 300°C, INJ = 300°C, OVEN = 150°C, *t_R* = 17.8 min for (*R*) and *t_R* = 17.2 min for (*S*). For 1-(3-methoxyphenyl)propanol: DET = 280°C, INJ = 280°C, OVEN = 120°C, *t_R* = 88.5 min for (*R*) and *t_R* = 89.2 min for (*S*). For 1-(4-methoxyphenyl)propanol: DET = 280°C, INJ = 280°C, OVEN = 130°C, *t_R* = 56.3 min for (*R*) and *t_R* = 57.2 min for (*S*). For 1-(4-fluorophenyl)propanol: DET = 280°C, INJ = 240°C, OVEN = 120°C, *t_R* = 23.6 min for (*R*) and *t_R* = 24.4 min for (*S*). For 1-(1-naphthyl)propanol: DET =

**Fig. 1.** Molecular structure of (1'*R*,2*R*,3*R*)-**L6a**.

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* Crystallographic data (CCDC 645974 and 645975) for the structures of **L6a** and **L7b** have been deposited with the Cambridge Crystallographic Data Centre. Copies of the data can be obtained, free of charge, on application to the Director, CCDC, 12 Union Road, Cambridge CB2 1EZ, United Kingdom (Fax: 44-1223-336033 or e-mail: deposit@ccdc.cam.ac.uk).

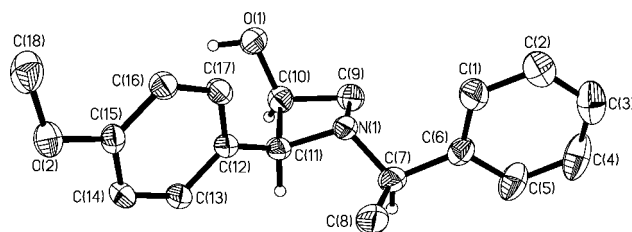


Fig. 2. Molecular structure of (1'*R*,2*S*,3*S*)-**L7b**.

280°C, INJ = 280°C, OVEN = 160°C, t_R = 66.3 min for (*R*) and t_R = 64.7 min for (*S*). The absolute configuration of the major enantiomer was assigned by the comparison of optical rotation with literature data.^{25,26}

RESULTS AND DISCUSSION

Toward the development of a chiral 3-hydroxyazetidine, we adopt (*R*)-1-phenylethylamine as the origin of chirality, which has been well employed to create a promising chiral reaction sphere.^{27,28} The chiral 3-hydroxyazetidines **L1a–L8b** are easily prepared by a three-step procedure (Scheme 1). Condensation of (*R*)-1-phenylethylamine with the appropriate aldehyde in dichloromethane in the presence of anhydrous $MgSO_4$ affords the corresponding aldimines **1**. Subsequently, performing the Staudinger reaction of imines **1** with acetoxyketene, derived from acetoxyacetyl chloride and Et_3N ,²⁹ gives a crude mixture of *cis*- β -lactams **2a** and **2b**. Finally, reduction of *cis*- β -lactams **2** (**2a** and **2b**) by AlH_2Cl , prepared from $LiAlH_4$ and $AlCl_3$ in ratio of 1:1 in dry THF,³⁰ gives a series of *cis*-3-hydroxyazetidines **L1a–8a** and **L1b–8b** in moderate to good overall yields after purification by flash chromatography (Table 1).

The configurations of aryl and hydroxyl groups in **L6a** and **L7b** are assigned as *cis* by their crystal structure (Figs. 1 and 2). The hydroxyl group and electron lone pair on N atom are situated on the same side of the azetidine backbone, indicating that a metal ion can easily chelate to both of them.

In order to obtain excellent enantioselectivity under mild conditions, the addition of diethylzinc to benzaldehyde using **L1a** has been studied at room temperature. Increasing the ratio of ligand **L1a** to benzaldehyde, increases the yield and enantioselectivity of (*S*)-1-phenyl-1-propanol (Table 2, entries 1–3), and the best enantioselectivity (94% ee) is obtained, when 20 mol% of **L1a** is used (Table 2, entry 3). A further increase in the ratio of ligand **L1a** to benzaldehyde does not increase the enantioselectivity (Table 2, entries 4 and 5). After having optimized the reaction conditions, various aromatic aldehydes have been studied for the ethylation using diethylzinc catalyzed by **L1a**. The additions to aromatic aldehydes provided the corresponding *S*-configuration alcohols as the major isomer with excellent enantioselectivities (79–97% ee) and in moderate to high yields (Table 2, entries 6–13) similar to the result obtained for benzaldehyde. Electron-deficient and electron-rich aromatic aldehydes are tolerated and exhibit similar enantioselectivities, and 1-naphthaldehyde gives the best enantioselectivity of 97% ee (Table 2, entry 13).

Ligands **L1a–8b** are roughly divided into two types, namely the *cis*-(1'*R*,2*R*,3*R*)-2-aryl-3-hydroxyazetidine ligands (**La**; **L1–8a**) and *cis*-(1'*R*,2*S*,3*S*)-2-aryl-3-hydroxyazetidine ligands (**Lb**; **L1–8b**). In exploring the addition of diethylzinc to aldehydes, ligands **L1a–8b** have been screened under the optimized conditions (20 mol % of ligand in toluene and 3 equiv of Et_2Zn at room temperature). Initial experiments showed that the addition reac-

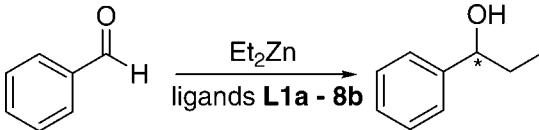
TABLE 2. Diethylzinc addition to aromatic aldehydes using ligand **L1a**^a

Entry	Ar'	Amount of ligand (mol%)	Yield ^b (%)	ee ^c (%) (config.)
1	Ph	5	36	87 (<i>S</i>)
2	Ph	10	53	91 (<i>S</i>)
3	Ph	20	76	94 (<i>S</i>)
4	Ph	30	85	94 (<i>S</i>)
5	Ph	50	96	94 (<i>S</i>)
6	2-MeOC ₆ H ₄	20	98	79 (<i>S</i>)
7	3-MeOC ₆ H ₄	20	43	93 (<i>S</i>)
8	4-MeOC ₆ H ₄	20	29	88 (<i>S</i>)
9	2-ClC ₆ H ₄	20	97	94 (<i>S</i>)
10	3-ClC ₆ H ₄	20	98	94 (<i>S</i>)
11	4-ClC ₆ H ₄	20	98	95 (<i>S</i>)
12	4-FC ₆ H ₄	20	80	91 (<i>S</i>)
13	1-naphthyl	20	74	97 (<i>S</i>)

^aReaction conditions: Et_2Zn (300 mol %), toluene as a solvent, at room temperature, 29 h.

^bIsolated yield after flash chromatography.

^cDetermined by GC analysis on a chiral cyclodextrin capillary column. Absolute configuration was determined by the comparison of specific rotation with literature data.^{25,26}

TABLE 3. Screening of ligands **L1a–8b** in asymmetric addition of diethylzinc to benzaldehyde^a


Entry	Ar (ligand)	Yield ^b (%)	ee ^c (%) (config.)
1	Ph (L1a)	76	94 (S)
2	2-ClC ₆ H ₄ (L2a)	78	83 (S)
3	3-ClC ₆ H ₄ (L3a)	54	86 (S)
4	4-ClC ₆ H ₄ (L4a)	48	73 (S)
5	2-MeOC ₆ H ₄ (L5a)	95	79 (S)
6	3-MeOC ₆ H ₄ (L6a)	82	89 (S)
7	4-MeOC ₆ H ₄ (L7a)	87	81 (S)
8	1-naphthyl (L8a)	98	93 (S)
9	Ph (L1b)	88	92 (R)
10	2-ClC ₆ H ₄ (L2b)	98	95 (R)
11	3-ClC ₆ H ₄ (L3b)	81	89 (R)
12	4-ClC ₆ H ₄ (L4b)	66	83 (R)
13	2-MeOC ₆ H ₄ (L5b)	98	57 (R)
14	3-MeOC ₆ H ₄ (L6b)	91	82 (R)
15	4-MeOC ₆ H ₄ (L7b)	88	90 (R)
16	1-naphthyl (L8b)	98	90 (R)

^aReaction conditions: Et₂Zn (300 mol %), ligand (20 mol %), toluene as a solvent, at room temperature, 29 h.

^bIsolated yield after flash chromatography.

^cDetermined by GC analysis on a chiral cyclodextrin capillary column. Absolute configuration was determined by the comparison of specific rotation with literature data.^{25,26}

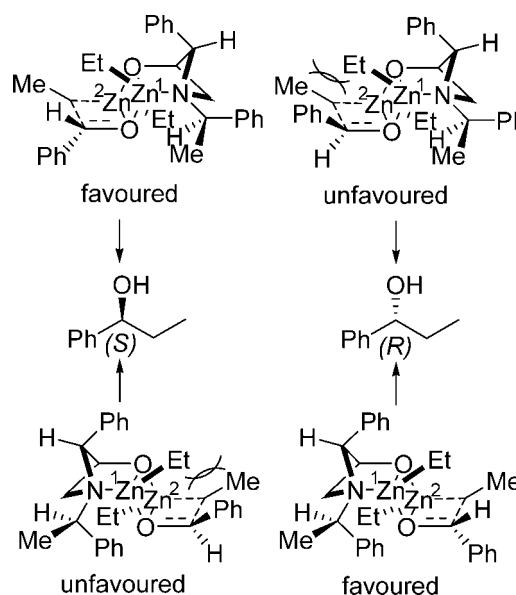
tion catalyzed by ligand **L1a** proceeded smoothly to afford (S)-1-phenyl-1-propanol in 76% yield and 94% ee (Table 3, entry 1). Similar to the results of **L1a**, all the other *cis*-(1*R*,2*R*,3*R*)-2-aryl-3-hydroxyazetidine ligands (**La**) induce enantiomeric excess in (S)-configuration alcohol, (S)-1-phenyl-1-propanol, as the major isomer with good to excellent enantioselectivities (73–93% ee) (Table 3, entries 2–8), while under the same condition, *cis*-(1*R*,2*S*,3*S*)-2-aryl-3-hydroxyazetidine ligands (**Lb**) induce the enantiomeric excess in (R)-configuration alcohol in good yields and enantioselectivities (Table 3, entries 9–16). The results of the reactions with ligands *cis*-(1*R*,2*R*,3*R*)-2-aryl-3-hydroxyazetidine ligands **La** can not be easily ruled out (Table 3, entries 1–8), while the results of the reactions with ligands **Lb** (**L1–8b**) demonstrate the noticeable steric and electronic effects of a variety of comparable substituents at the *ortho*, *meta*, and *para* positions of the phenyl group. For example, moving chlorine atom from *para* to *meta* to *ortho* position results in increased enantiomeric excess (Table 3, entries 10–12) due to steric effects, and the best enantioselectivity (95% ee) is obtained by using **L2b** as catalyst (Table 3, entry 10). In contrast, moving MeO group from *para* to *meta* to *ortho* position results in decreased enantiomeric excess (Table 3, entries 13–15) due to the coordination effects of MeO group, and the lowest enantioselectivity (57% ee) is obtained by using **L5b** as catalyst (Table 3,

entry 13). Replacement of the hydrogen atom at *para* position (**L1b**) with an electron-withdrawing group (chloro, **L4b**) results in decreased yield and enantiomeric excess (Table 3, entry 12), while electron-rich ligands **L7b** with an electron-donating group (MeO) at *para* position and **L8b** with a 1-naphthyl group exhibit similar activity and selectivity as ligand **L1b** does (Table 3, entries 9, 15 and 16).

The stereochemical outcome of these diethylzinc additions to aldehydes is in agreement with the *anti*-transition state structures proposed by Noyori and coworkers,^{31–35} which may well explain our experimental observations that the substituted aryl groups on the azetidine backbone have considerable influence on the enantioselectivity of the ligands **L1a–8b** and the configuration of the resulting adducts (Fig. 3). The reasons for the different stereoelectronic sensitivity between ligands, **La** and **Lb**, are not clear at this time, however, the sterically encumbered environment around the metal center seems to be a major reason for these differences. Further computational investigation of the ligand architecture for this transformation is currently underway.

CONCLUSIONS

In conclusion, a new series of chiral *cis*-3-hydroxyazetidines have been synthesized from (R)-1-phenylethylamine. These ligands promote the addition of diethylzinc reagent to aromatic aldehydes under mild conditions to afford the corresponding secondary alcohols in high yield with good to excellent enantioselectivity. These results demonstrate that the chiral azetidine-based ligand is promising in asymmetric catalysis and deserves certainly more attention. Further efforts will focus on the applications of these ligands toward other types of asymmetric reactions and the exploration of new ligands based on chiral azetidine.

**Fig. 3.** Proposed transition states.

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